MakeUsSparkle

End-to-End System Health & Technical Audit Report

Prepared For: Executive Leadership & Client
Audit Date: October 2026
Status: Action Required Before Public Scale

EXECUTIVE SUMMARY FOR BUSINESS STAKEHOLDERS

Your platform has a **gorgeous visual design** and rich specialized features for personalized jewelry (such as custom charm bracelets, gift hampers, PhonePe checkout, and iThink shipping). Customers will love the aesthetic and customization options.

However, a technical audit of the underlying engine revealed **critical vulnerabilities in pricing, payments, and inventory**. Currently, the system trusts the web browser to dictate product prices, does not automatically record payments if a user closes their browser, never reduces inventory stock when products are purchased, and allows single-use coupons to be reused indefinitely.

This document explains these issues in **simple, everyday terms with real-world examples** and provides an immediate roadmap to make the store completely safe, robust, and profitable.

At-A-Glance System Health Scorecard

Area	Status	What This Means for Your Business
Storefront & Custom Charm UX	READY & IMPRESSIVE	Visually stunning, mobile-responsive, and offers seamless personalized jewelry selection.
Order Pricing & Payment Safety	CRITICAL RISK	The server accepts payment amounts sent by the user's browser without verifying against the database.
Inventory & Stock Management	CRITICAL RISK	Stock numbers are never deducted after sales; risk of selling items you do not have in stock.
Coupons & Discounts	HIGH RISK	Single-use promo codes never increment their usage counter, allowing unlimited reuse.
Payment Confirmation (PhonePe)	HIGH RISK	Missing automated server webhook; orders may remain "pending" if the customer closes their app.
Shipping Logistics (iThink)	WELL INTEGRATED	Live rates, automatic AWB generation, and courier tracking are cleanly implemented.
Cart Customization Memory	CUSTOMER UX BUG	Adding two different charm designs of the same product overwrites the first design in the cart.

The 5 Critical Issues Explained in Plain English

1. The "Self-Discounting" Price Loophole (Client-Trusted Pricing)

CRITICAL FINANCIAL RISK

REAL-WORLD EVERYDAY EXAMPLE

> Imagine a customer walks into your physical jewelry store, picks up a necklace marked **₹2,499**, scratches out the price tag with a pen, writes **₹1** on it, and walks up to the cash register. Your cashier blindly accepts the ₹1 without looking up the item on the official price sheet.

WHAT HAPPENS IN THE CODE

When a customer clicks "Pay", their browser sends the total price to the backend server. The server simply forwards that number directly to PhonePe without recalculating the true item prices from the database. A tech-savvy user can modify this number to ₹1 in their browser console.

BUSINESS IMPACT & FIX

> **Impact:** Massive risk of revenue loss and malicious exploitation.

> **The Fix:** The backend must recalculate the total price itself by fetching the verified price of every product and charm directly from the database before generating any payment link.

2. Phantom Inventory (Products Never Run Out of Stock)

OPERATIONAL & FULFILLMENT RISK

REAL-WORLD EVERYDAY EXAMPLE

> You have exactly **5 pieces** of a special gold bracelet in stock. Over the weekend, 25 different customers purchase that bracelet on the website. Because the system never subtracts 1 each time a piece is bought, it continues to show "5 in stock" forever.

WHAT HAPPENS IN THE CODE

While your products have a "stock" field in the database, the order creation and payment confirmation routines never decrement this number.

BUSINESS IMPACT & FIX

> **Impact:** You end up taking money for items you cannot deliver, resulting in customer complaints, negative reviews, and manual refund processing.

> **The Fix:** Add an automatic stock decrement (e.g. `stock = stock − quantity`) the second an order is confirmed, and reject checkout if stock reaches zero.

3. The Disappearing Payment Link (Missing PhonePe Webhook)

CUSTOMER SUPPORT HEADACHES

REAL-WORLD EVERYDAY EXAMPLE

A customer pays ₹1,500 on PhonePe via UPI on their mobile phone. The money leaves their bank account successfully. But right before their browser redirects back to your website's "Thank You" screen, their battery dies, or they close the tab. PhonePe has the money, but your website never receives the confirmation.

WHAT HAPPENS IN THE CODE

The system currently relies 100% on the customer's browser landing on `/payment/status` to verify the transaction. There is no automated server-to-server webhook callback from PhonePe.

BUSINESS IMPACT & FIX

Impact: The order stays marked as "Pending" in your admin panel. The customer calls customer support complaining: *"My money was deducted, but I didn't get an order confirmation!"*

The Fix: Implement a PhonePe Webhook listener. PhonePe will directly notify your server in the background, even if the customer's phone switches off.

4. Unlimited Single-Use Coupons

MARKETING MARGIN LOSS

REAL-WORLD EVERYDAY EXAMPLE

> You give an influencer a one-time promo code **"SAVE500"** meant for 1 order only. The customer uses it on Monday. Then they come back on Tuesday, Wednesday, and Thursday and use it again every single day because the coupon is never punched or stamped as used.

WHAT HAPPENS IN THE CODE

The database model contains a `usedCount` counter, but the order and payment services never update or increment this counter when an order is completed.

BUSINESS IMPACT & FIX

> **Impact:** Profit margins are eroded when users share one-time promotional codes on discount forums.

> **The Fix:** Automatically increment `usedCount` by 1 upon payment verification and prevent checkout once the limit is reached.

5. Custom Charm Memory Mix-up in the Shopping Cart

CUSTOM JEWELRY FULFILLMENT ERROR

REAL-WORLD EVERYDAY EXAMPLE

> A customer designs a custom charm bracelet with charms **"M - O - M"** for Mother's Day. She loves it, so she clicks to design a second bracelet of the exact same style, but with charms **"D - A - D"** for Father's Day. She opens her cart and finds the cart only has two "M - O - M" bracelets—her second custom design was completely wiped out!

WHAT HAPPENS IN THE CODE

In the cart logic, when checking if an item is already in the cart, it only checks the base product ID and color. It fails to check if the charms or custom messages are different, so it simply changes the quantity from 1 to 2 instead of adding a new line item.

BUSINESS IMPACT & FIX

> **Impact:** High customer frustration and making the wrong custom jewelry items in the workshop.

> **The Fix:** Update the shopping cart to consider unique charm selections when grouping items.

Recommended Step-by-Step Action Plan

All of these issues are straightforward to resolve with standard software engineering practices. Here is the suggested priority plan:

1. **Immediate Financial Protection (1–2 Days)**
   Implement server-side price recalculation on order creation so no customer can alter prices in the browser.

2. **Inventory & Stock Decrements (1 Day)**
   Connect order placement to stock deduction so products accurately display out-of-stock when inventory reaches zero.

3. **PhonePe Background Webhook (1–2 Days)**
   Add a dedicated server-to-server webhook endpoint so orders are marked "paid" even if customers close their screens.

4. **Coupon Usage & Cart Customization Fix (1 Day)**
   Increment coupon usage counters upon order payment, and allow multiple distinct custom charm designs in the cart.

5. **Speed & Cloudinary Optimization (1–2 Days)**
   Turn on Cloudinary's automatic format and quality compression (WebP/AVIF) to make the store load 3x faster on mobile phones.